

Motivation

Trade of indivisible goods between unit-demand agents with money.
Introduced by Quinzii (1984), Gale (1984), and Svensson (1984).



- House allocation problem without money solved by Gale's Top Trading Cycle algorithm.
- Housing market with **quasilinear utilities** (in money) solved, e.g., by Hungarian algorithm.

For high-value items (e.g., houses):

- agents experience income effects,
- willingness to pay depends on level of wealth.

Nothing known about computational complexity of 'more realistic' assumptions, e.g. 'soft' budget constraints or costs of borrowing.

Gale's Housing Market

- Agents $\{1, \dots, n\}$ and houses $\{1, \dots, n\}$.
- Agent i has **preference sets** $P_0^i, P_1^i, \dots, P_n^i$ covering $\mathbb{R}^n$.
- Demands house j at prices $\mathbf{p} \in P_j^i$, and **nothing** at $\mathbf{p} \in P_0^i$.
- Indifferent between houses j and k if $\mathbf{p} \in P_j^i \cap P_k^i$.

competitive equilibrium: Prices $\mathbf{p} \in \mathbb{R}^n$ and envy-free allocation $\pi : [n] \rightarrow [n]$ of houses to agents: $\mathbf{p} \in P_{\pi(i)}^i$.

ε -**approximate competitive equilibrium:** Prices $\mathbf{p} \in \mathbb{R}^n$ and allocation $\pi : [n] \rightarrow [n]$ of houses to agents s.t. $\mathbf{p}$ is ε -close to $P_{\pi(i)}^i$.

Gale's assumptions

- P_j^i are closed.
- For each house j : $\mathbf{p} \notin P_j^i$ if $p_j \geq 1$.
- $P_1^i, \dots, P_n^i$ cover lower faces of n -dimensional unit cube.

Theorem: (Gale, 1984) Competitive equilibrium exists if agents satisfy Gale's assumptions.

Theorem: ε -approximate competitive equilibrium exists if agents satisfy Gale's assumptions (ii) and (iii).

Computational Questions

How hard is it to compute (approximate) competitive equilibria?

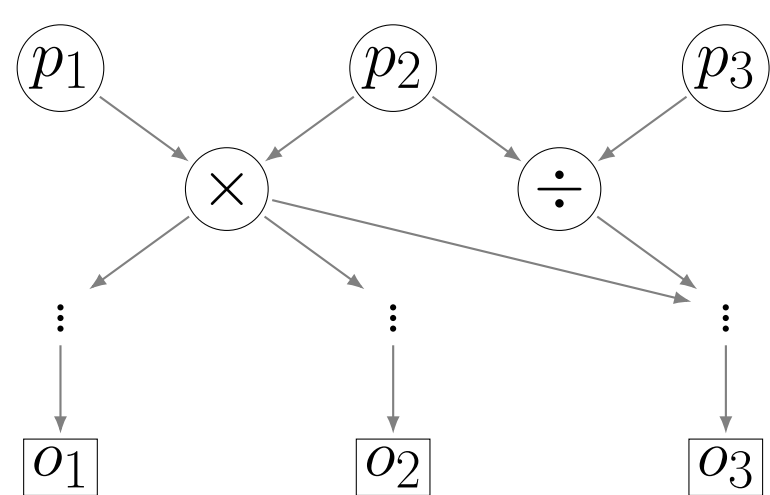
Housing: Given agent preferences that satisfy Gale's assumptions, compute equilibrium $(\mathbf{p}, \pi)$.

ε -Housing: Given $\varepsilon > 0$ and agent preferences that satisfy Gale's assumptions (ii) and (iii), compute ε -equilibrium $(\mathbf{p}, \pi)$.

Computational representation of preference sets: $P_0, P_1, \dots, P_n$

Arithmetic circuits

- Input: prices $\mathbf{p} \in \mathbb{R}^n$.
- Output: $\mathbf{o} \in \mathbb{R}^{n+1}$ with $o_j = 0$ iff $\mathbf{p} \in P_j$.
- Base $\{+, -, \times, \div, \min, \max\}$, rational constants.



Polynomial-time algorithms

- Input: prices $\mathbf{p} \in \mathbb{R}^n$.
- Output: Houses demanded at $\mathbf{p}$.
- Polynomial runtime guarantees.

```
26 function unit_demand(p: Dict{Int, Int})
27   max_utility = 0
28   Ψ = Set{Int}()
29   for u ∈ intersect(keys(p), trades)
30     u = util(u, Set{u})
31     if u > max_utility
32       max_utility = u
33       Ψ = Set{u}
34   end
35   end
36   return Ψ
```

Function oracles $f_1, \dots, f_n$.

- Input: prices $\mathbf{p} \in \mathbb{R}^n$.
- Output: $f_j(\mathbf{p}) = 1$ iff $\mathbf{p} \in P_j$.
- Similar to demand queries in auctions



Access Our Paper



Illustration with Two Houses

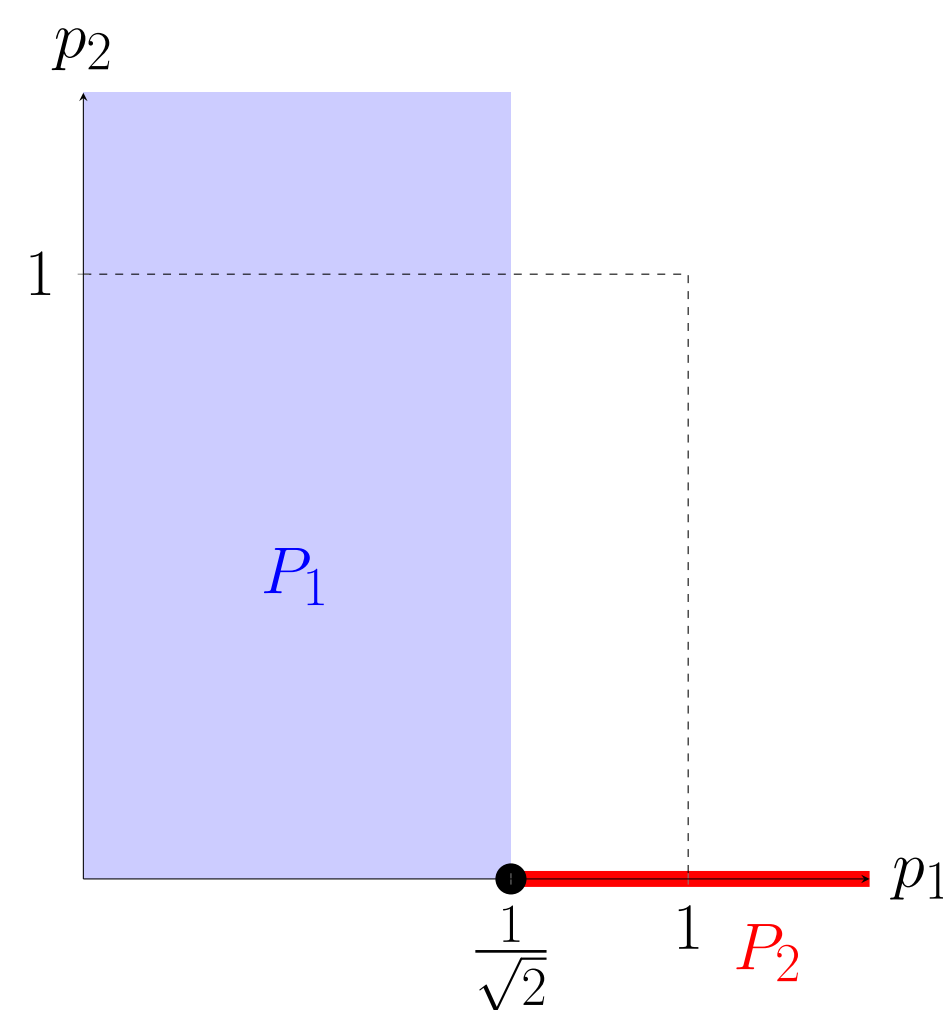
Two agents with identical preferences

$P_0 = \mathbb{R}^n$.

$P_1 = \{\mathbf{p} \in \mathbb{R}^n \mid 0 \leq p_1 \leq \frac{1}{\sqrt{2}}\}$.

$P_2 = \{\mathbf{p} \in \mathbb{R}^n \mid p_1 \geq \frac{1}{\sqrt{2}} \text{ and } p_2 = 0\}$.

Unique equilibrium prices at $\mathbf{p} = (\frac{1}{\sqrt{2}}, 0)$



Results

How hard is it to find a competitive equilibrium?

Theorem 1: Finding an exact competitive equilibrium in the circuit model is FIXP-complete.

Theorem 2: Finding an approximate equilibrium in the algorithm or query model takes a polynomial number of queries / steps if the market consists of two agents.

Theorem 3: Finding an approximate equilibrium in the algorithm model is PPAD-complete, even if we restrict the market to three identical agents.

Theorem 4: Finding an approximate equilibrium in the query model takes exponentially many query in the approximation parameter for four or more agents.

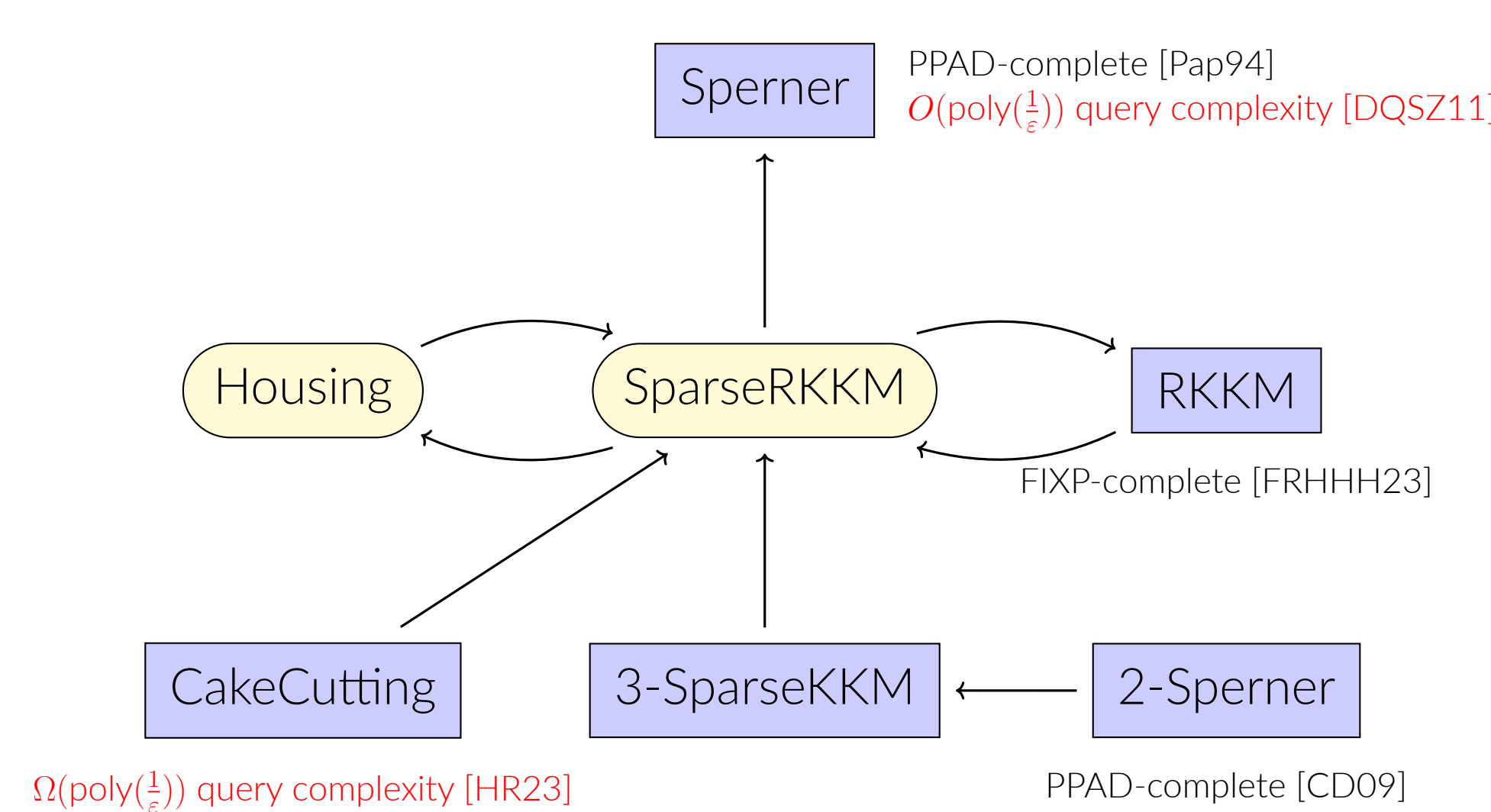
Implication: We can't find competitive equilibria efficiently.

Complexity Classes

PPAD: approximate Nash equilibrium, approximate market equilibria in Arrow-Debreu and Fisher markets, cake cutting, Sperner, End-of-the-Line.

FIXP: (Exact) Nash equilibrium, KKM, Rainbow-KKM, etc.

Proof Overview



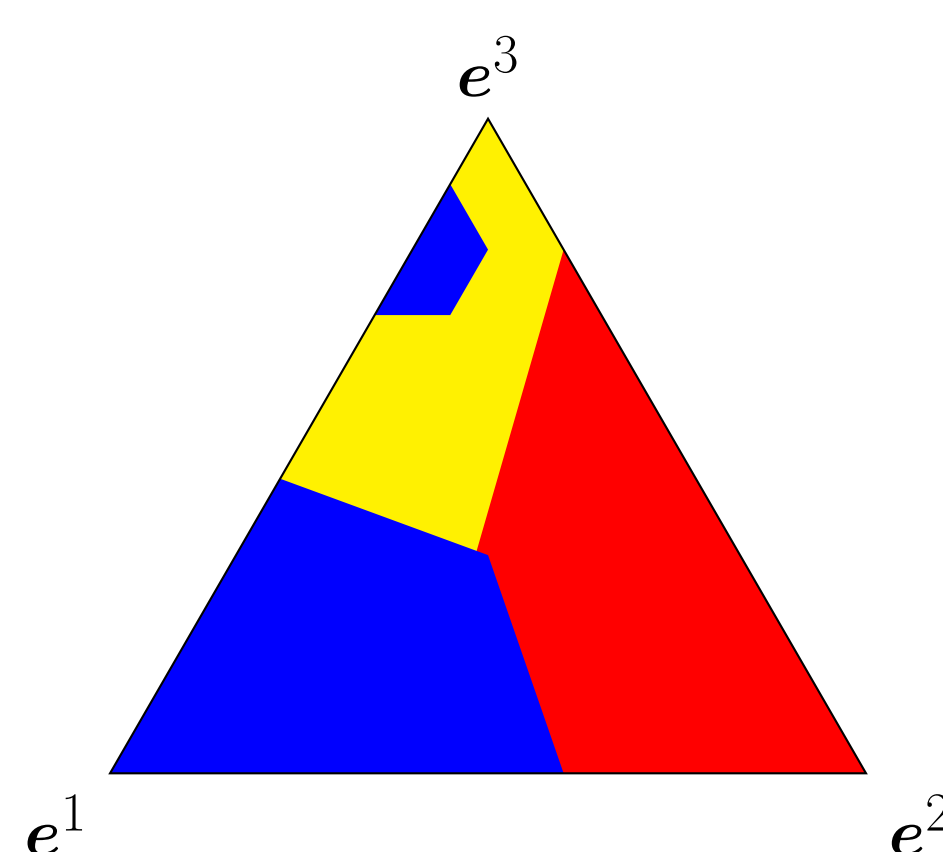
(Rainbow-)KKM

Standard simplex $\Delta_{n-1} := \text{conv}\{\mathbf{e}^j \mid j \in [n]\}$.

F_S is the face of Δ_{n-1} spanned by vertices $\{\mathbf{e}^j \mid j \in S\}$.

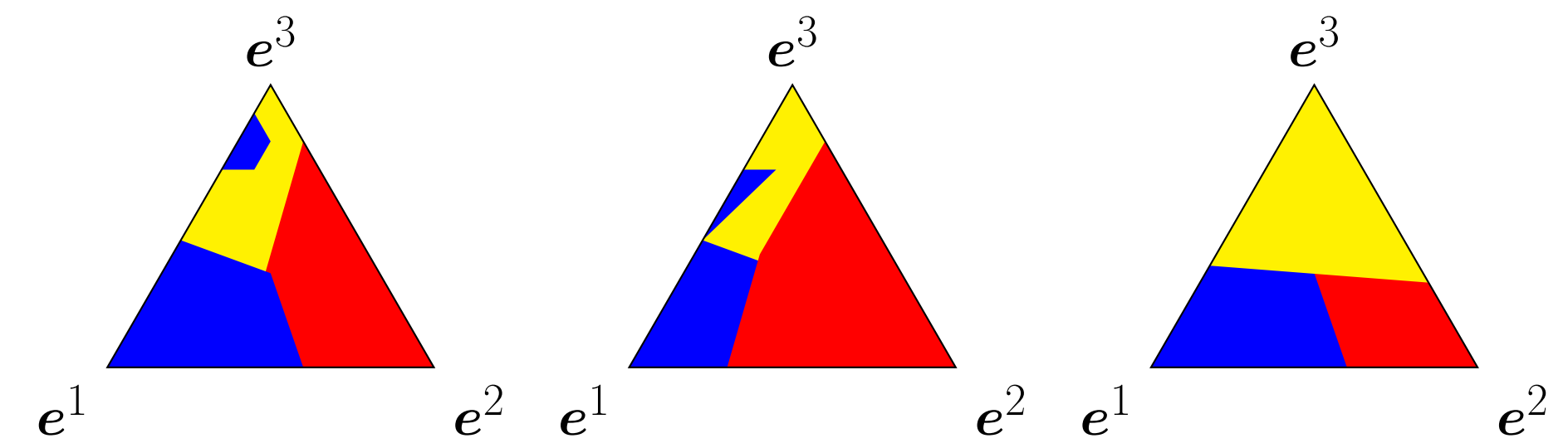
KKM covering: $C_1, \dots, C_n \subseteq \Delta_{n-1}$ s.t. $F_S \subseteq \bigcup_{j \in S} C_j$ for all $S \subseteq [n]$.

Sparse KKM covering: For every $j \in [n]$, $\mathbf{x} \in C_j$ implies $x_j > 0$.



(Approximate) KKM Lemma (cf. Knaster et al., 1929):

- For any KKM covering $C_1, \dots, C_n$ of closed sets, there exists $\mathbf{x}$ contained in every C_j .
- For any KKM covering $C_1, \dots, C_n$ and $\varepsilon > 0$, there exists $\mathbf{x}$ ε -close to each C_j .



(Approximate) Rainbow-KKM Lemma (cf. Gale, 1984):

- Any collection of n KKM coverings $C_1^i, \dots, C_n^i$ with closed sets admits a point $\mathbf{x}$ and permutation π of $[n]$ so that $\mathbf{x}$ lies in every $C_{\pi(i)}^i$.
- Any collection of n KKM coverings $C_1^i, \dots, C_n^i$, admits a point $\mathbf{x}$ and permutation π of $[n]$ so that $\mathbf{x}$ is ε -close to every $C_{\pi(i)}^i$.

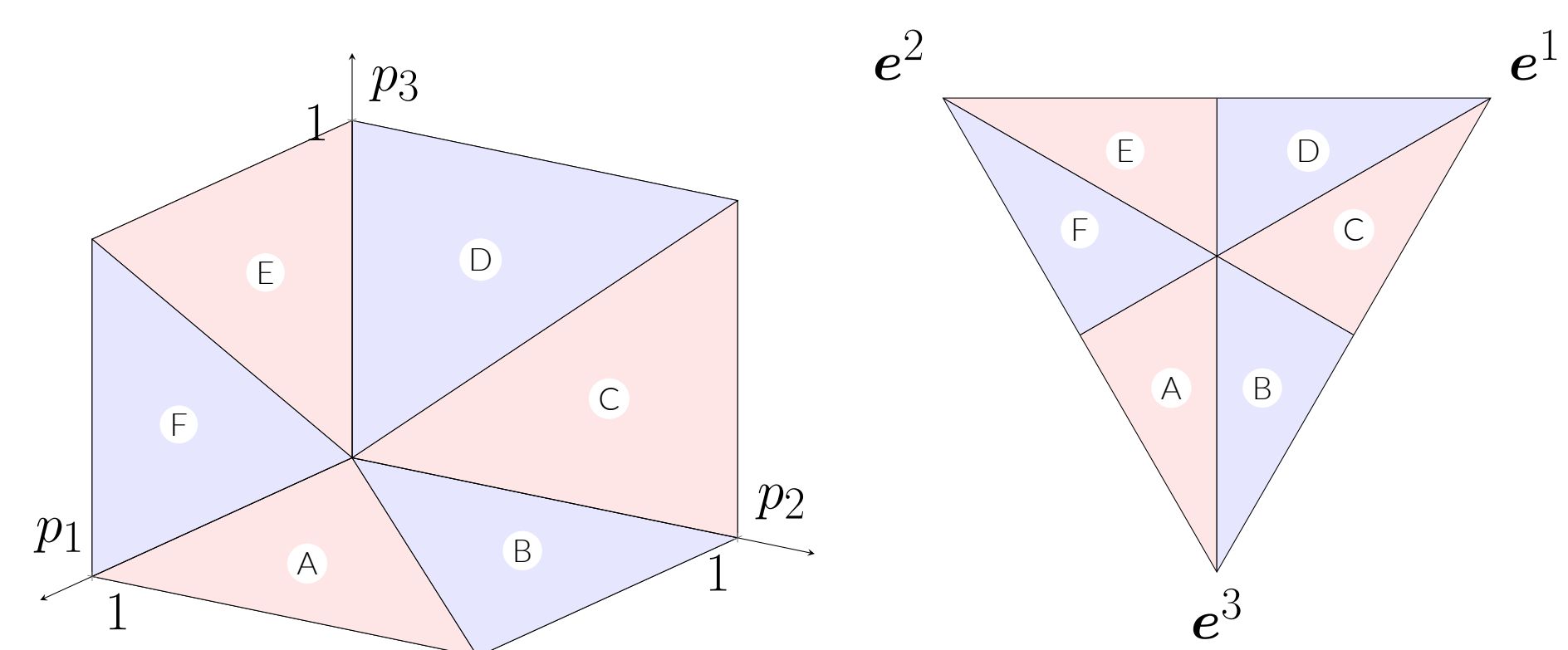
The Equivalence of Housing and Sparse RKKM

Gale (1984) defines **homeomorphism** ϕ .

Divide domain of housing market into $n!$ simplices corresponding to permutations π of $[n]$.

$$\Sigma_\pi := \{\mathbf{p} \in \Sigma_n \mid p_{\pi(1)} \geq p_{\pi(2)} \geq \dots \geq p_{\pi(n)} = 0\}$$

$$\phi(\mathbf{p})_{\pi(k)} := \frac{1 - p_{\pi(1)}}{n} + \frac{p_{\pi(1)} - p_{\pi(2)}}{n-1} + \frac{p_{\pi(k-1)} - p_{\pi(k)}}{n-k+1}.$$

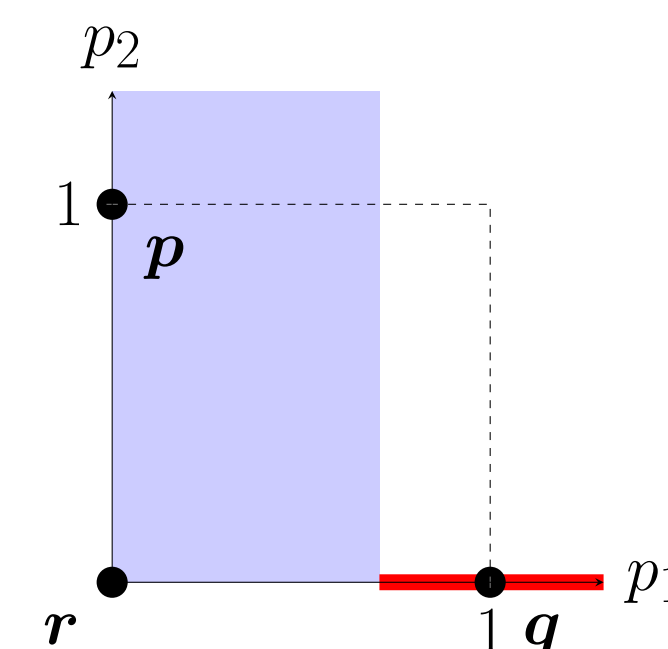


The Reductions

	Housing		SparseRKKM
Instance	$(P_1^i, \dots, P_n^i)_{i \in [n]}$	$\Rightarrow$	$(\phi(P_1^i), \dots, \phi(P_n^i))_{i \in [n]}$
Solution	$\phi^{-1}(\mathbf{x}), \pi$	$\Leftarrow$	$\mathbf{x}, \pi$

Lemma: ϕ is n -Lipschitz and ϕ^{-1} is n^2 -Lipschitz, which gives reductions for approximation versions.

Two-Agent Markets



Algorithm 1 Binary search
Let $\mathbf{p} \leftarrow (0, 1) \in C_1^1$, $\mathbf{q} \leftarrow (1, 0) \in C_2^2$ and $\mathbf{r} \leftarrow (0, 0)$.
while $\|\mathbf{p} - \mathbf{q}\| > \varepsilon$ and $\mathbf{r} \in C_1^1 \cup C_2^2$ **do**
 Update $\mathbf{p} \leftarrow \mathbf{r}$ if $\mathbf{r} \in C_1^1$ and $\mathbf{q} \leftarrow \mathbf{r}$ else.
 $\mathbf{r} \leftarrow \frac{1}{2}(\mathbf{p} + \mathbf{q})$.
return ε -equilibrium prices $\mathbf{r}$.

Outlook

We initiate the study of housing markets from the perspective of computational complexity.

We hope these results will stimulate further examination of the complexity of markets with income effects:

- What is the complexity of Housing with utility functions instead of preference sets?
- Does hardness continue to hold under the natural assumptions of monotonicity in money and no externalities as in (Quinzii, 1984; Svensson, 1984)?
- How might CakeCutting with monotonic valuations (Deng et al., 2012; Hollender & Rubinstein, 2023) map to assumptions for Housing?

References

- [CD09] Xi Chen and Xiaotie Deng, *On the complexity of 2D discrete fixed point problem*, Theoretical Computer Science **410** (2009), no. 44, 4448–4456.
- [DQS211] Xiaotie Deng, Qi Qi, Amin Saberi, and Jie Zhang, *Discrete fixed points: Models, complexities, and applications*, Mathematics of Operations Research **36** (2011), no. 4, 636–652.
- [FRHH23] Aris Filos-Ratsikas, Kristoffer A Hansen, Kasper Høgh, and Alexandros Hollender, *Fixp-membership via convex optimization: Games, cakes, and markets*, SIAM Journal on Computing (2023), FOCSS21–30.
- [Gal84] David Gale, *Equilibrium in a discrete exchange economy with money*, International Journal of Game Theory **13** (1984), no. 1, 61–64.
- [HR23] Alexandros Hollender and Aviad Rubinstein, *Envy-free cake-cutting for four agents*, 2023 IEEE 64th Annual Symposium on Foundations of Computer Science (FOCS), IEEE, 2023, pp. 113–122.
- [Pap94] Christos H Papadimitriou, *On the complexity of the parity argument and other inefficient proofs of existence*, Journal of Computer and system Sciences **48** (1994), no. 3, 498–532.
- [Qui84] Martine Quinzii, *Core and competitive equilibria with indivisibilities*, International Journal of Game Theory **13** (1984), no. 1, 41–60.
- [Sve84] Lars-Gunnar Svensson, *Competitive equilibria with indivisible goods*, Zeitschrift für Nationalökonomie/Journal of Economics **44** (1984), no. 4, 373–386.